

Oscillations in a charged cylinder — Solution

X22 (2022) — English translation

A1

0.50 pts

A disk with radius R is charged with surface charge density σ_R . Determine the potential $\varphi(y)$ at a point on the axis at a distance y from the center of the disk. The potential is zero at infinity.

Potential from a ring of width dr with radius r

$$d\varphi = k \frac{\sigma_R \cdot 2\pi r dr}{\sqrt{y^2 + r^2}}$$

Potential from all disk find integration:

$$\varphi(y) = \pi k \sigma_R \int_0^R \frac{d(r^2)}{\sqrt{y^2 + r^2}} = 2\pi k \sigma_R \left(\sqrt{R^2 + y^2} - |y| \right)$$

Answer:

$$\varphi(y) = 2\pi k \sigma_R \left(\sqrt{R^2 + y^2} - |y| \right)$$

A2

1.00 pts

Two such disks of radius R charged with surface charge density $\sigma_R > 0$ are parallel to each other. The distance between the centers of the disks is $2L$, the centers are located on the axis of the disks. In the equilibrium position there is a charge q with mass m , which can only move along the axis of the disks. Determine the angular frequency ω_1 of vibrations of such a charge. What is the charge sign?

When the charge is displaced by x along the axis, an outer ring of width $\delta = 2Rx/L$ appears on the nearest disk, which creates a restoring force. The total field of the remaining charges is zero. The field of this ring is

$$E_x(x) = -k \frac{2\pi R \delta \sigma_R}{(L^2 + R^2)} \frac{L}{\sqrt{L^2 + R^2}} = -\frac{\sigma_R R^2}{\varepsilon_0 (L^2 + R^2)^{3/2}} x.$$

Equation of motion charge

$$m\ddot{x} = qE(x).$$

Angular frequency for oscillation:

$$\omega^2 = \frac{q\sigma_R R^2}{m\varepsilon_0 (L^2 + R^2)^{3/2}}.$$

Answer:

$$\omega^2 = \frac{q\sigma_R R^2}{m\varepsilon_0 (L^2 + R^2)^{3/2}},$$

$$q > 0.$$

A3

1.00 pts

Now this charge can only move in a perpendicular direction. Express the angular frequency ω_2 of oscillations in this case in terms of ω_1 . What is the charge sign now?

We use Gauss's theorem for a small cylinder between the disks to find the dependence of the radial field $E_r(r)$ on the displacement from the axis r :

$$2\pi r \cdot 2xE_r(r) + 2\pi r^2 E_x(x) = 0$$

$$E_r(r) = -\frac{1}{2}E_x(x)\frac{r}{x}.$$

Charge must be other sign. It can be seen, that coefficient in linear dependence is 2 times less, therefore,

$$\omega_2 = \frac{\omega_1}{\sqrt{2}}.$$

Answer:

$$\omega_2 = \frac{\omega_1}{\sqrt{2}},$$

$$q < 0.$$

B1

1.00 pts

The lateral surface of a cylinder with radius R and length L is charged with surface charge density σ_L . Determine the potential at a point on the axis at a distance z from the center of one of the bases of the cylinder. The potential is zero at infinity.

Potential from a ring of height dl at a distance l from zero

$$d\varphi(y) = k \frac{\sigma_L \cdot 2\pi R dl}{\sqrt{(z+l)^2 + R^2}}.$$

Potential from all side surface cylinder find integration:

$$\varphi(y) = 2\pi k R \sigma_L \int_z^{L+z} \frac{d(l+z)}{\sqrt{(l+z)^2 + R^2}} = 2\pi k R \sigma_L \left(\operatorname{arth} \frac{L+z}{\sqrt{(L+z)^2 + R^2}} - \operatorname{arth} \frac{z}{\sqrt{z^2 + R^2}} \right).$$

Answer:

$$\varphi(y) = 2\pi k R \sigma_L \left(\operatorname{arth} \frac{L+z}{\sqrt{(L+z)^2 + R^2}} - \operatorname{arth} \frac{z}{\sqrt{z^2 + R^2}} \right).$$

B2

1.00 pts

Two such cylinders (with radius R and length L , surface charged with surface charge density $\sigma_L > 0$) are placed close together and have a common axis. In the equilibrium position there is a charge q with mass m , which can only move along the axis of the cylinders. Determine the angular frequency ω_3 of vibrations of such a charge. What is the charge sign?

When the charge is displaced by x along the axis, an outer ring of width $\delta = 2x$ appears on the far cylinder, which creates a restoring force. The total field of the remaining charges is zero. The field of this ring is

$$E_x(x) = k \frac{2\pi R \delta \sigma_L}{(L^2 + R^2)} \frac{L}{\sqrt{L^2 + R^2}} = \frac{\sigma_L R L}{\varepsilon_0 (L^2 + R^2)^{3/2}} x.$$

Equation of motion charge

$$m\ddot{x} = qE_x(x),$$

it can be seen, that charge must be negative. Angular frequency for oscillation:

$$\omega_3^2 = \frac{|q|\sigma_L R L}{m\varepsilon_0 (L^2 + R^2)^{3/2}}.$$

Answer:

$$\omega_3^2 = \frac{|q|\sigma_L R L}{m\varepsilon_0 (L^2 + R^2)^{3/2}},$$

$$q < 0.$$

B3

0.50 pts

Now this charge can only move in a perpendicular direction. Express the angular frequency ω_4 of oscillations in this case in terms of ω_3 . What is the charge sign now?

Everything happens similarly to A3, that is,

$$E_r(r) = -\frac{1}{2}E_x(x)\frac{r}{x},$$

means $\omega_4^2 = \omega_3^2/2$ and the charge should also change sign.

Answer:

$$\omega_4 = \frac{\omega_3}{\sqrt{2}},$$

$$q > 0.$$

C1

1.50 pts

A charged cylinder with radius R and height $L = 40R/9$ consists of a side surface and one base. Surface charge density of the side surface σ_L , base σ_R . If you place a point charge at the center of the opposite base, it will be in an equilibrium position. Define the relation σ_L/σ_R .

The field strength created by the base of the cylinder will be found as

$$E_R = - \left. \frac{d\varphi}{dy} \right|_L.$$

This intensity equal

$$E_R = 2\pi\sigma_R k \left(1 - \frac{L}{\sqrt{R^2 + L^2}} \right).$$

Intensity field, created lateral wall cylinder find as

$$E_L = - \left. \frac{d\varphi}{dz} \right|_0.$$

This intensity equal

$$E_L = 2\pi\sigma_L k \left(1 - \frac{R}{\sqrt{R^2 + L^2}} \right).$$

Total field equal zero, hence obtain:

$$\frac{\sigma_L}{\sigma_R} = - \frac{\sqrt{R^2 + L^2} - L}{\sqrt{R^2 + L^2} - R} = - \frac{41 - 40}{41 - 9} = - \frac{1}{32}$$

Answer:

$$\frac{\sigma_L}{\sigma_R} = - \frac{\sqrt{R^2 + L^2} - L}{\sqrt{R^2 + L^2} - R} = - \frac{1}{32}$$

C2

2.50 pts

A charged cylinder with radius $R = 28b$ and height $L = 45b$ consists of a side surface and one base. Side surface charge $\sigma_L = -8\sigma_0$, base charge $\sigma_R = 25\sigma_0 > 0$. A particle with charge $q > 0$ is placed on the axis of this system. Estimate numerically the z coordinates (in b units) of the equilibrium positions if the particle can only move along the axis. Coordinate z is calculated as in the picture. Do this as accurately as possible, however, 1% accuracy is sufficient. Answers falling within 1% of correct will receive full marks.

We are looking for a point at which the total field is equal to zero. Next, we measure all lengths in units of b . For points with coordinates $z > -45$ we obtain the equation

$$E_{base}(z) + E_{side}(z) = 0,$$

$$25 \left(\frac{45+z}{\sqrt{28^2 + (45+z)^2}} - 1 \right) - 8 \left(\frac{28}{\sqrt{28^2 + (45+z)^2}} - \frac{28}{\sqrt{28^2 + z^2}} \right) = 0.$$

So that solve equation on calculator, representable it in form $z = f(z)$:

$$f(z) = -\frac{901}{25} + \sqrt{28^2 + (45+z)^2} \left(1 - \frac{224}{25\sqrt{28^2 + z^2}} \right).$$

We obtain answer $z = 0$. One can it substitute in $f(z)z < -45$, we compose following equation (change field disk):

$$25 \left(\frac{45+z}{\sqrt{28^2 + (45+z)^2}} + 1 \right) - 8 \left(\frac{28}{\sqrt{28^2 + (45+z)^2}} - \frac{28}{\sqrt{28^2 + z^2}} \right) = 0.$$

So that solve equation on calculator, representable it in form $z = f(z)$:

$$f(z) = -\frac{901}{25} - \sqrt{28^2 + (45+z)^2} \left(1 + \frac{224}{25\sqrt{28^2 + z^2}} \right).$$

Given equation converges very slowly. One can find it root method binary search: substitute some number and carrying out iterations observe, increase or decrease quantity z . Correct answer $z = -1632.60163579414519440461610908047948157856626649004333963591...$

Answer:

$$\frac{z}{b} = 0.$$

$$\frac{z}{b} = -1632.60163579414519440461610908047948157856626649004333963591...$$

C3

1.00 pts

Under the conditions of the previous paragraph, the particle was placed in the equilibrium position closest to the cylinder, its mass m . Determine the angular frequency ω of small vibrations of the particle.

Select a point with coordinate $z = 0$. We expand the field strength and find that at this point $E(z) = -\alpha z$, $\alpha = 2\pi\sigma_0 k \cdot \frac{560}{2809} \frac{z}{b}$. From the vibration equation

$$m\ddot{z} = -q\alpha z$$

we get the answer.

Answer:

$$\omega^2 = \frac{560\sigma_0}{5618\varepsilon_0 b m}$$